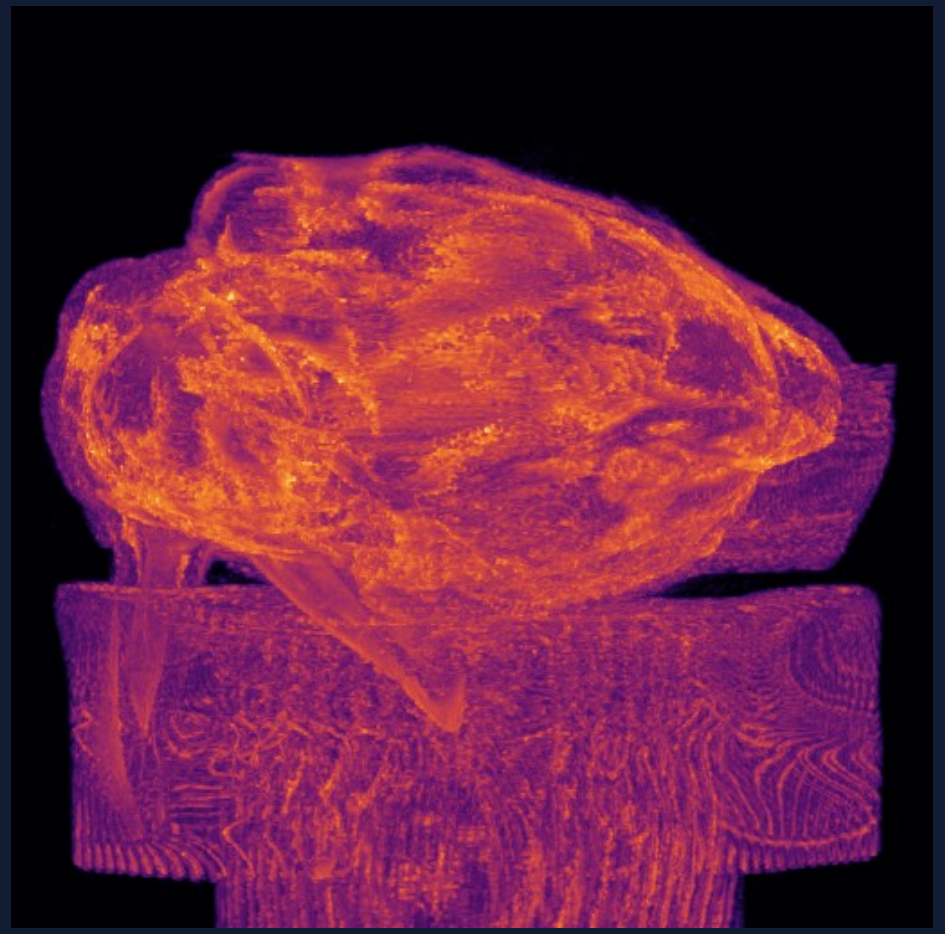
LARGE-SCALE COMPUTING · PRACTICAL ASSIGNMENT 3

BunnyMIP

Volumetric Maximum Intensity Projection with CUDA

A GPU port of a three-stage volumetric rendering pipeline — thresholding · 3×3×3 Gaussian blur · rotated MIP — validated against a CPU gold standard and benchmarked.

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Maximum Intensity Projection of the Stanford Bunny CT volume

Three stages over a CT volume

Stanford Bunny CT 512 × 512 × 361 uint16 · ≈ 180 MiB · 9.5×10^7 voxels **CPU baseline ≈ 6.8 s → port to the GPU**

1 THRESHOLD

Point operation

$v < T \Rightarrow 0$

remove air noise

2 GAUSSIAN BLUR

3×3×3 stencil

weighted average of

27 neighbours

3 MIP

Ray reduction

max intensity along

each rotated ray

DATA LAYOUT contiguous 1-D uint16 array · $idx = z \cdot N^2 + y \cdot N + x$ (N = 512, x fastest-varying)

The same threshold, σ and rotation matrix R feed both the CPU and GPU pipelines — making the validation meaningful.

The CUDA kernels — point & stencil

THRESHOLD

POINT OP

`256 thr/block · grid-stride`

- **Embarrassingly parallel** 1 thread ↔ many voxels
- **In place** v < T ⇒ 0, on d_volume
- **Fully coalesced** consecutive threads → addresses

```
if (v[i] < T) v[i] = 0;
```

GAUSSIAN BLUR

STENCIL

`8×8×8 blocks · out-of-place`

- **27-tap convolution** weights in constant memory
- **Clamped boundaries** mirrors the host exactly
- **__restrict__ + cast** (uint16_t) truncation

```
out = Σ  c_weights[k] · in[clamp(n)]
```

The CUDA kernels — the MIP reduction

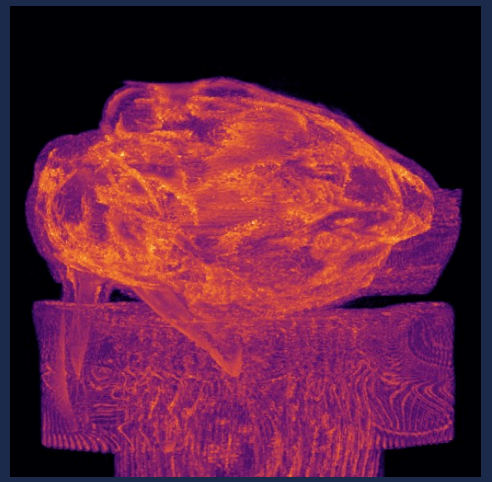
MAXIMUM INTENSITY PROJECTION

RAY-CAST

`16×16 blocks · one thread per output pixel`

- **Backward mapping** screen → volume via c_R, nearest-neighbour
- **810-step ray** ray_range = 405 (half-diagonal) → full diagonal
- **__ldg read-only cache** the volume is never written by this kernel
- **Private maximum** no atomics, no synchronisation

One ray per pixel → max intensity

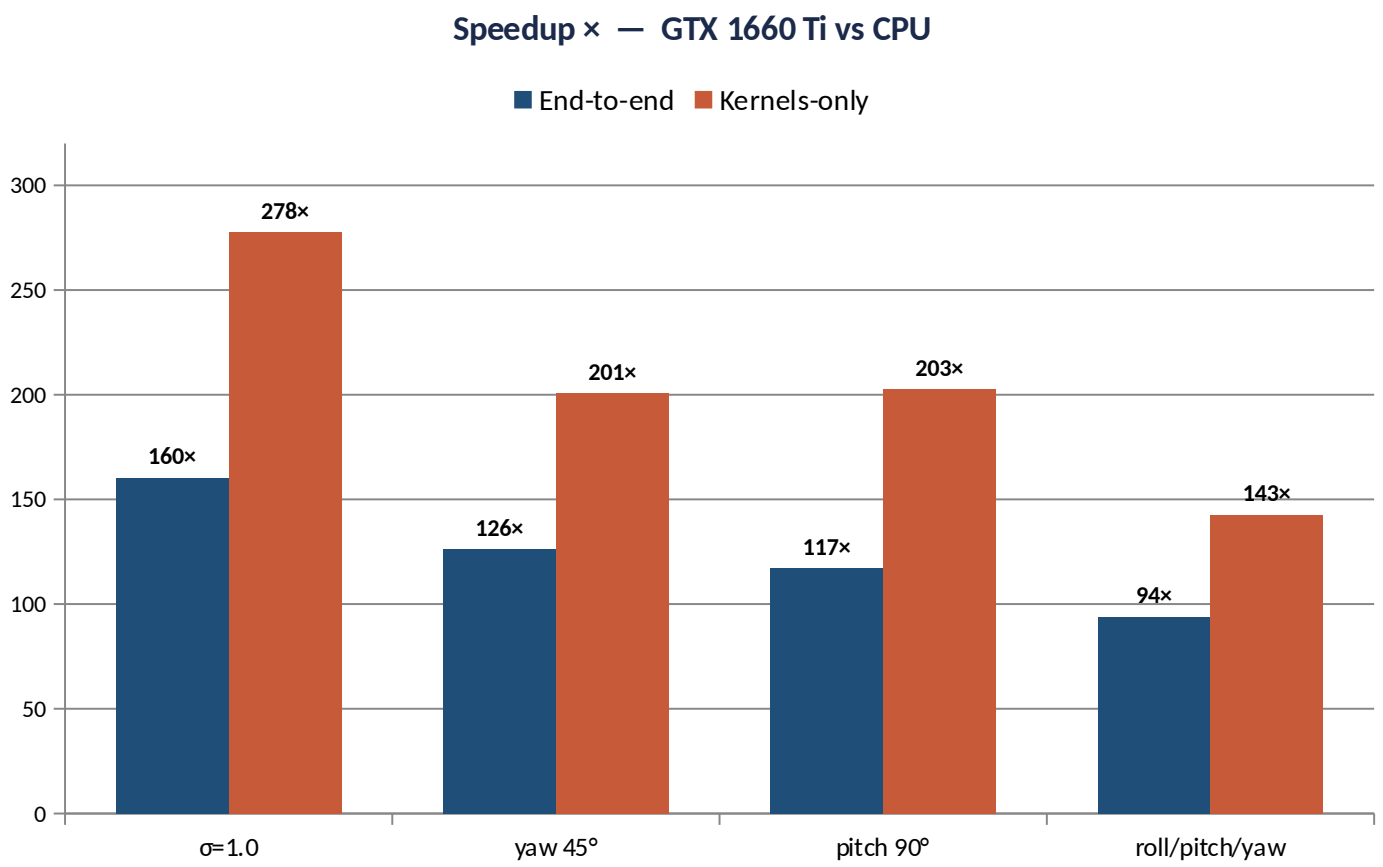


the projected raster (d_raster)

BY CONSTRUCTION

No locks · no atomics · no race conditions. Read-only inputs in constant memory; the volume is read through the read-only cache.

Bit-stable — and 90–290× faster



SPEEDUP

93–165× end-to-end

143–287× kernels

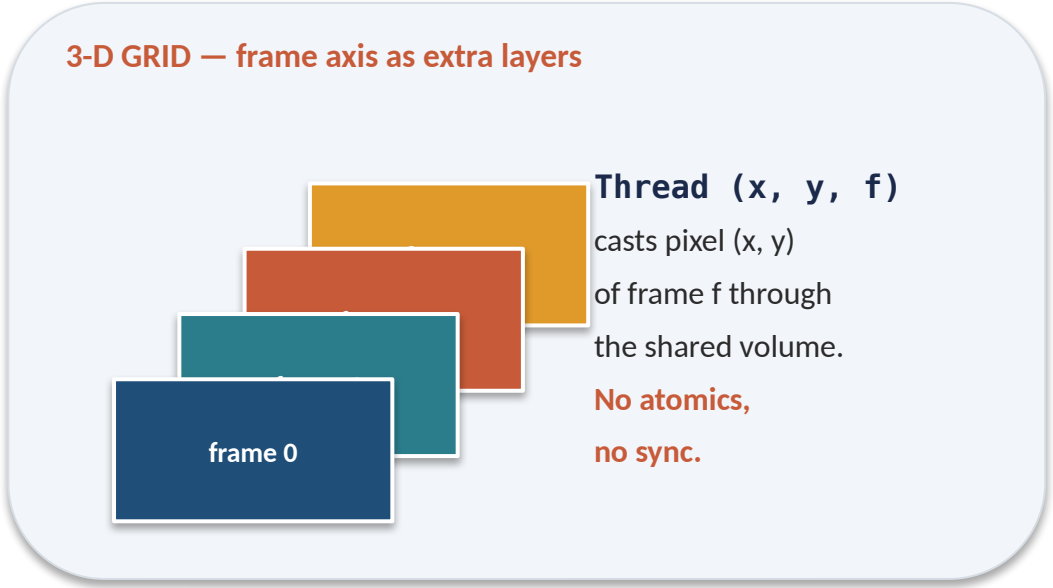
VALIDATION

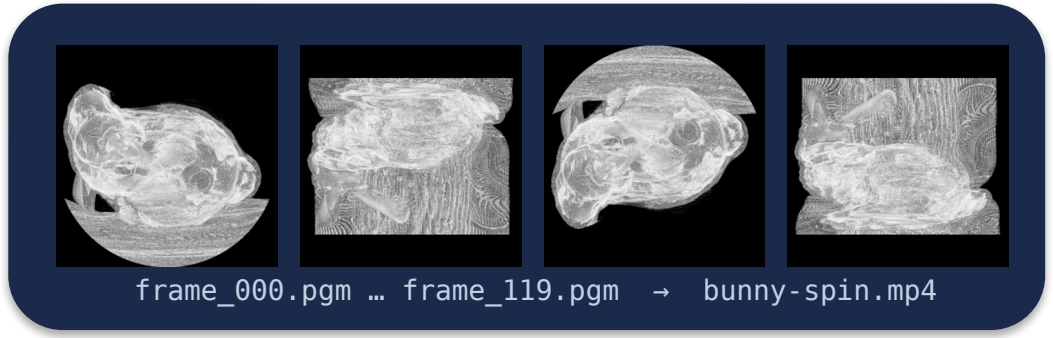
0.00 % differing

- **max abs diff = 0** bit-stable vs CPU gold standard
- **PASS** tolerance ≤ 1.0 %

N MIP frames in a single kernel call

- **Threshold + blur run once** the blurred volume is reused for every frame
- **3-D grid (x, y, frame)** grid z-dimension — one layer per frame
- **One launch renders all N MIPs** each frame uses its own c_Rmulti[f]
- **Encoded into a rotating video** MAX_FRAMES = 512 · FFmpeg, normalised to 8-bit





SUMMARY

Three kernels, one validated GPU pipeline

Faithful port

Bit-stable with the CPU gold standard — 0.00 % difference

Large speedup

93–165× end-to-end, up to 287× kernels-only

Sound CUDA design

Constant memory · grid-stride · __ldg · no atomics

Innovation

N MIP frames in a single kernel call → rotating video